Jin Hyuk Kim

M.S. Student in Computer Engineering

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Research Interests

Interests — AI4Science for Drug Discovery, Molecular Generation, Agentic AI, Molecular Representation Learning, Explainable/Interpretable AI

Education

2025.03 – Present — **Hallym University**
M.S. in Computer Engineering
AI-Powered Cheminformatics Laboratory (APC Lab) (Advisor: Jonghwan Choi)

2019.03 – 2025.02 — **Hallym University**
B.S. in Big Data, Division of Software / Double Major in Smart IoT
Capstone Project: CanChem

Publications

* *Corresponding author*

International Journal

1. **Jin Hyuk Kim**, Gyeong Hwan Kim, Hyeon Jun Park, Jonghwan Choi*, "CUE: A Chemical Uncertainty-Aware Embedding Framework for Multi-modal Drug Selectivity Prediction," 2026. (**On Revision**)
2. **Jin Hyuk Kim**, Jonghwan Choi*, "OCSAug: Diffusion-based Optical Chemical Structure Data Augmentation for Improved Hand-drawn Chemical Structure Image Recognition," *The Journal of Supercomputing*, 2025.

International Conference

1. **Jin Hyuk Kim**, Jonghwan Choi*, "Agentic Multi-objective Molecular Optimization via Dynamic Routing of Property-specific Editor Networks," in *AI4Sci Conference Korea*, 2026. (**Accepted**)

Domestic Journal

1. Tae Woong Song**, Jin Hyuk Kim**, Hyun Jun Park, Jonghwan Choi*, "A Study on Semantic Molecular Prompt Engineering Method for Large Language Model-based Molecular Property Prediction," *KIISE Transactions on Computing Practices*, 2026.
2. Tae Woong Song**, Jin Hyuk Kim**, Hyun Jun Park, Jonghwan Choi*, "Pretrained Large Language Model-based Drug-Target Binding Affinity Prediction for Mutated Proteins," *Journal of KIISE*, 2025.

Domestic Conference

1. Hyun Jun Park**, Jin Hyuk Kim**, Tae Woong Song, Jonghwan Choi*, "M-WPDTA: Mutation-Aware Drug-Target Affinity Prediction with Wildtype Referenced Dual-Encoder Modeling," in *Korea Computer Congress* (*KCC*), Poster, 2025.
2. Tae Woong Song**, Jin Hyuk Kim**, Hyun Jun Park, Jonghwan Choi*, "A Study on Molecular Prompt Engineering Method for Large Language Model-based Molecular Property Prediction," in *Korea Computer Congress* (*KCC*), Poster, Outstanding Presentation Paper Award, 2025.
3. **Jin Hyuk Kim**, Tae Woong Song, Jonghwan Choi*, "A Study on DDPM-based Molecular Generation and Semi-Supervised Learning for Improving the Performance of Optical Chemical Structure Recognition," in *Annual Symposium of KIPS* (*ASK*), Oral, 2024.

Teaching Experience

Jul. 2025 — **Teaching Assistant, AI-Driven Drug Discovery Program for High School Students**
Hallym University, supported by the Gangwon State Office of Education

Spring 2025 — **Teaching Assistant, Reinforcement Learning**
Hallym University

Fall 2024 — **Teaching Assistant, Database Basics**
Hallym University

Additional Information

Patents

2025.04.28 — **키나아제에 대한 화합물의 선택성을 예측하는 전자 장치의 동작 방법**
최종환, **김진혁**
대한민국 특허출원 10-2025-0055457

사전학습된 거대 언어 모델을 이용하여 돌연변이 단백질에 대한 약물의 결합 친화도를 예측하는 전자 장치, 방법 및 프로그램
최종환, 송태웅, **김진혁**, 박현준
대한민국 특허출원 10-2025-0055458

Honors & Awards

2026.05 — **Outstanding Graduate Research Paper Award**
Presented at the 44th Anniversary Ceremony of Hallym University

Certifications

2026.06 — **ADsP: Advanced Data Analytics Semi-Professional**
Korea Data Agency